Answers to this paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allotted for writing the answers

***Section A** is compulsory. Attempt* **any four** *questions from* **Section B.**
The intended marks for questions or parts of questions are given in the brackets.

SECTION –A

*(Attempt **all** questions in this section)*

Question 1 **(15)**

Choose the correct answers to the questions from the given options.
(Do not copy the question, write the correct answers only.)

i. 5 kWh can be expressed as
a) 3.6×10^6 J
b) 1.8×10^7 J
c) 3×10^5 J
d) Cannot be expressed in joule as it is a unit of power

ii. In which of the following examples is heat energy converted to electric energy
a) Steam engine
b) Electromagnet
c) Geyser
d) Thermocouple

iii. Compare uniform circular motion and uniform linear motion. Identify the statement that is true.
a) Both have uniform velocity
b) Uniform linear motion is accelerated motion
c) Uniform circular motion has a constant speed but a variable velocity
d) Uniform linear motion has varying speed but constant velocity

iv. A truck of 5×10^3 kgf and a cart weighing 500 kgf are moving at the same speed. Compare their kinetic energies.
a) 1:1
b) 10:1
c) 1:10
d) 5:1

v. Statement (A): Out of alpha, beta and gamma radiations, alpha particles have the maximum penetration power.
Statement (B): The alpha particles are the heaviest of the three.
a) Both A and B are true and B is the correct explanation of A.
b) Both A and B are true but B is not the correct explanation of A.
c) A is false but B is true.
d) A is true B is false.

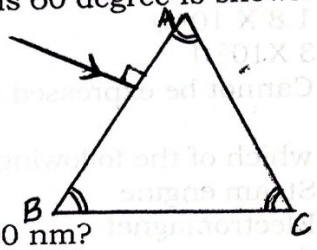
vi. The IR spectrum of radiations and the UV spectrum of radiations can be obtained with the help of prisms made by the following material, respectively.
a) Rock salt, Glass
b) Quartz, Glass
c) Rock salt, Quartz
d) Quartz, Rock salt

vii. Pick out the statement that is TRUE for refractive index of glass with respect to air
a) is the ratio of wavelength of light in glass to the wavelength of light in air
b) the ratio of the sine of the angle of incidence to the sine of the angle of refraction
c) is 1.33
d) the ratio of the speed of light in glass to the speed of light in air

viii. A body is acted upon by two unequal forces in opposite directions, but not in the same line. The effect is that
a) The body will only have rotational motion
b) The body will only have translational motion
c) The body will have neither rotational motion nor translational motion
d) The body will have both rotational motion and translational motion

ix. Select the incorrect answer.
a) An actual machine always has efficiency less than 100 per cent
b) The mechanical advantage of a machine can be less than 1
c) A machine can be used as a speed multiplier and force multiplier at the same time
d) A machine can have mechanical advantage less than its velocity ratio

x. An equilateral prism of a material whose critical angle is 60 degree is shown in the DIAGRAM.A ray is incident on the side AB.
What is the angle of incidence on surface AC?
a) 90°
b) 0°
c) 60°
d) 30°



xi. Identify which radiation has a wavelength less than 800 nm?
a) Red light
b) Blue
c) Microwaves
d) Radio waves

xii. Identify the true statement.
a) The deviation of red is less than yellow for a given pair of media.
b) The critical angle of violet is more than red for a given light and pair of media.
c) For glass, the refractive index of blue light is more than the refractive index of red light.
d) The wavelength of blue is more than yellow.

xiii. In the block and tackle pulley of 5 pulleys,if the tensions in the string are represented as "T", how will the Load (L) and effort (E) relate to the tension (T)?
a) L= E=T
b) L=4 T, E =4 T
c) L= 5 T, E=T
d) L =4 T, E =T

xiv. A bulb is rated 100W-220V.It is being used at 110V supply. The power consumed by the bulb is:
(a) 100W
(b) half of 100W
(c) nearly 25W
(d) none of the above

xv. Which of the following is not true for an electromagnet?
a) It is made of soft iron.
b) Its magnetic field strength can be changed.
c) The polarity of the electromagnet cannot be changed
d) It can be easily demagnetised

Question 2

Complete the following : **(12)**

a) When we are exposed to radioactive substances such as K^{40}, C^{14} and radium present inside our bodies, we say we are exposed to _____ radiation. (4)

b) When the stem of a tuning fork is pressed on a table, the table top starts vibrating. These vibrations are an example of _____ vibrations.

c) 5 bulbs are connected in series in a room. One bulb fuses. It is removed and the remaining 4 bulbs are again connected in series in the same circuit. The resistance will now_____ and the intensity of the bulbs will_____.

d) An electric kettle is rated 3kW, 250V. Can this device be used in a circuit with a fuse of current rating 13 A.?(Yes/No)

ii. Name the following: (4)

a) The process which requires very high temperature and high pressure for two light nuclei to form a heavy nucleus by combination.

b) The heat energy required to melt unit mass of ice at 0°C to water at 0°C without a change in temperature.

c) The application of the echo method used to detect the presence of an obstacle and also find its range with a signal of e.m radiation.

d) The type of vibration where restoring force and resistive force of the surrounding medium causes the amplitude of motion to decrease.

iii. Why do farmers fill their fields with water on a cold winter night? (2)

iv. Name which wire the following are connected in

a) Switch

b) Fuse (2)

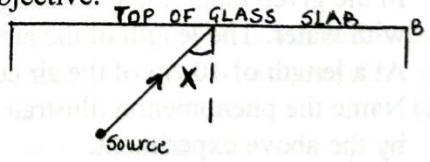
Question 3 **(13)**

i. What are two advantages of an electromagnet over a permanent magnet? (2)

ii. A scientist while performing an experiment desires that no incident light escapes from the top of a glass block, AB. He varies the angle "X" shown in the DIAGRAM continuously and finally manages to achieve his objective. (3)

a) What is angle "X"?

b) Explain why he was able to achieve his goal.



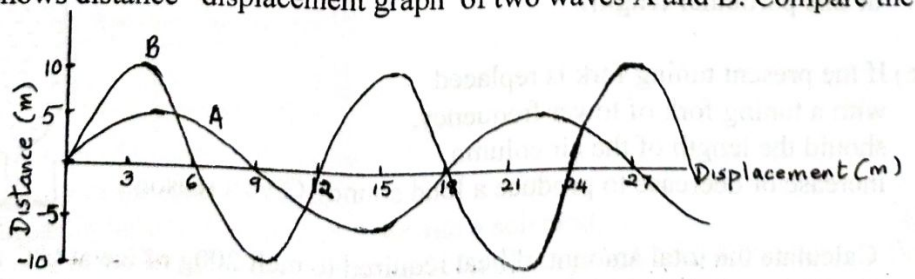
iii. Give one use of

a) an electromagnet

b) Gamma radiations of $_{27}Co^{60}$.

c) special film badges used by those who work with radioactive materials. (3)

iv. The figure shows distance –displacement graph of two waves A and B. Compare the



a) Amplitude

b) Wavelength of the two waves. (2)

v. Give one reason

a) Why a lower note is obtained on a flute when some of the holes are closed?

b) Why is the loudness of a plucked wire increased when the wire is mounted on a sound board? (2)

vi. A fuse is made with an alloy of _____ and _____. (1)

SECTION B

(Attempt **any four** questions in this section)

Question 4

i. Give the class of lever to which each of the following belongs:

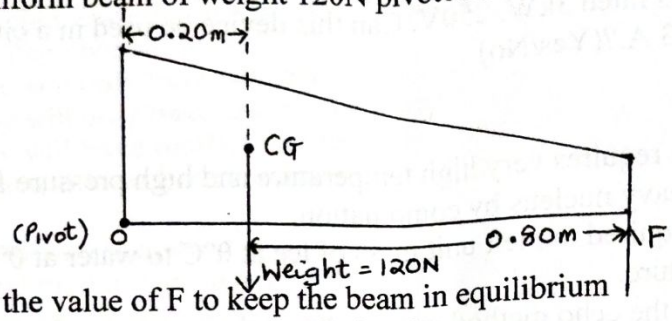
a) Wheelbarrow

b) Lemon squeezer

c) Physical balance

d) Nut cracker (4)

ii. A non- uniform beam of weight 120N pivoted at one end , **O** is shown in the diagram below.



a) Calculate the value of F to keep the beam in equilibrium

b) When is a body said to be in equilibrium?

c) What order of lever is the beam? (3)

iii. The given table gives the refractive indices of some substances.

Rock salt	Diamond	Water	Turpentine	Kerosene oil
1.56	2.41	1.33	1.47	1.41

a) Arrange these materials in ascending order of optical density.

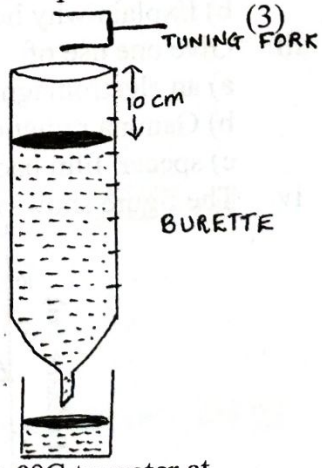
b) In which medium does light travel the fastest?

c) Out of the following pairs of media, which pair exhibits total internal reflection of light?

Kerosene oil to rock salt, Diamond to water, Kerosene oil to Turpentine, Water to kerosene oil (3)

Question 5

i. In the given diagram, a vibrating tuning fork is kept near the mouth of a burette filled with water. The length of the air column is adjusted by opening the tap of the burette. At a length of 10 cm of the air column, a loud sound is heard. (3)



a) Name the phenomenon illustrated by the above experiment.

b) Why is a loud sound heard at this particular length?

c) If the present tuning fork is replaced with a tuning fork of lower frequency, should the length of the air column increase or decrease to produce a loud sound? Give a reason.

ii. Calculate the total amount of heat required to melt 200g of ice at 0°C to water at 100°C.(Specific latent heat of ice 336J/g, specific heat capacity of water= 4.2J/g/°C. (3)

iii. Write the nuclear equations for the following nuclear changes: (3)

${}_{Z}X^{A}$ emits a beta particle to become X_1, X_1 emits a gamma radiation to become X2, X_2 then it emits an alpha radiation to become X3.

$${}_{Z}X^{A} \xrightarrow{\beta} X_1 \xrightarrow{\gamma} X_2 \xrightarrow{\alpha} X_3$$

iv. Mention one similarity between nuclear fusion and nuclear fission. (1)

Question 6

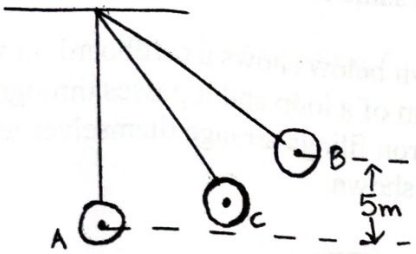
A hollow ice cream cone has a height of 12 cm

a) Where is the position of its centre of gravity from the broad base?

b) Will the position change if it is filled completely with honey? (Yes /No) (2)

ii. The image of a candle flame placed at a distance of 36cm from a spherical lens is formed on a screen placed at a distance of 72 cm from the lens. Calculate the focal length of the lens and its power. (3)

iii. The figure alongside shows a simple pendulum of mass 200g.It is displaced from the mean position A to the extreme position B. The potential energy at position A is zero. At position B the pendulum bob is raised by 5m. (3)

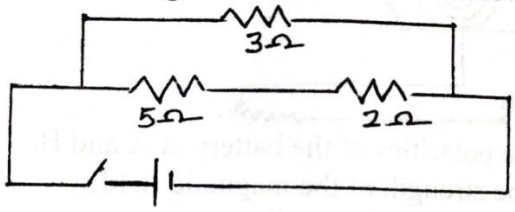


a) What is the potential energy of the bob at position B?

b) What is the total mechanical energy at point C?

c) What is the speed of the bob at position A when released from B?

iv. Observe the given circuit diagram and answer the following questions: (2)



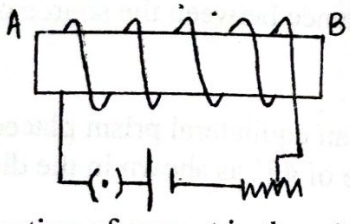
a) Calculate the resistance when the key K completes the circuit.

b) Calculate the current through the 2Ω resistance when the circuit is complete

Question 7

i. A bird is flying at a height of 3m above the river surface while a fish is 4m below the surface. At what depth would the fish appear to the bird? Take R.I of water with respect to air =4/3. (2)

ii. The diagram below shows an electromagnet with a soft iron core.



a) Mark the direction of current in the wire.

b) Mark the polarity at the ends A and B.

c) Draw the magnetic field lines inside and outside a solenoid. (3)

iii. A cube of ice of mass 30g at 0°C is added to 200g of water at 30°C. Calculate the final temperature of water after the whole ice cube has melted. Specific Latent heat of ice = 80 cal/g; Specific heat capacity of water =1cal/g ◦C (2)

iv. Give reason for the following. (2)

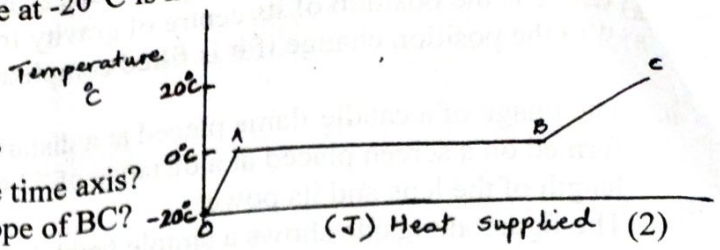
a) Ice cream seems colder in the mouth than water at 0°C.

b) The colour of the sky in a direction other than the sun is blue.

v. Mention one safety precaution while storing radioactive material. (1)

Question 8

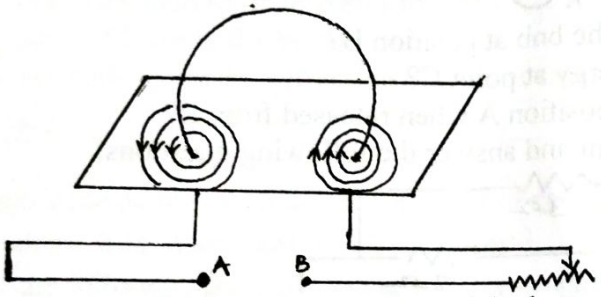
i. In the GRAPH, a given quantity of ice at -20°C is heated. (3)



a) What does the portion AB represent?
b) Why is the portion AB parallel to the time axis?
c) Why is the slope OA steeper than slope of BC? (2)

ii. State
a) One condition required for work done by a force to be zero.
b) One feature that is the same for the nuclei of all isotopes of Uranium -235.

iii. The DIAGRAM shown below shows a cardboard on which iron filings are placed. A wire is bent in the form of a loop and it passes through the cardboard. When the current flows through it, the iron filings arrange themselves as shown and the direction of the magnetic field is also shown.



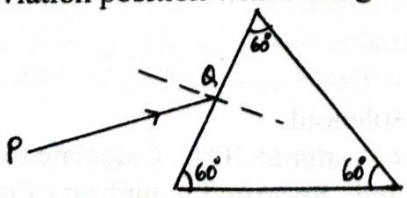
a) Draw the figure and state the polarities of the battery at A and B.
b) State one way to decrease the strength of the magnetic field.
c) State one way to change the polarity of the coil. (3)

iv. An object AB is placed in front of a concave lens between F_2 and $2F_2$ on the principal axis. Use the standard rays to obtain the position of the image. (2)

Question 9

i. A man standing between two cliffs produces a sound and hears two successive echoes at intervals of 3s and 4s respectively.
a) Calculate the distance between the two cliffs. The speed of sound in air is 330m/s.
b) Why will an echo not be heard when the distance between the source of sound and the reflecting surface is 10m? (3)

ii. A monochromatic ray of light is incident on an equilateral prism placed at a minimum deviation position with an angle of incidence of 45° as shown in the diagram. (3)



a) Copy the diagram and complete the path of the ray PQ
b) State two factors on which angle of deviation depends.

iii. Name the following: (2)
a) The two wires that are the same potential at the local substation.
b) The material used to make a calorimeter.

iv. a)Write the nuclear fission equation for Uranium 235 (2)